SAME SIZE ARTWORK
LEAFLET SIZE : 115 mm x 280 mm

50 mm

35 mm

SPACE FOR PHARMACODE

For the use only of a Registered Medical Practitioners or a Hospital or a Laboratory

Ezzicad™ 10 mg Tablets

(Ezetimibe 10 mg)

Label Claim
Each uncoated tablet contains:
Ezetimibe............10 mg.

PHARMACEUTICAL FORM
Tablets
White to off-white, capsule shaped, flat, beveled edged uncoated tablets engraved with glenmark logo 'G' on one side and'44'on the other side.

CLINICAL PARTICULARS

Therapeutic indications

Primary Hypercholesterolaemia
Ezetimibe, administered alone, or with an HMG-CoA reductase inhibitor (statin), is indicated as adjunctive therapy to diet for use in patients with primary (heterozygous familial and non-familial) hypercholesterolaemia.

Ezetimibe, administered in combination with fenofibrate, is indicated as adjunctive therapy to diet for the reduction of elevated total-C, LDL-C, Apo B, and non-HDL-C in patients with mixed hyperlipidemia.

Prevention of Cardiovascular Events
Ezetimibe, administered with a statin, is indicated to reduce the risk of cardiovascular events (cardiovascular death, nonfatal myocardial infarction, nonfatal stroke, hospitalization for unstable angina, or need for revascularization), in patients with coronary heart disease (CHD) and a history of acute coronary syndrome (ACS).

Homozygous Familial Hypercholesterolaemia (HoFH)
Ezetimibe, administered with a statin, is indicated for patients with HoFH. Patients may also receive adjunctive treatments (e.g. LDL apheresis).

Homozygous Sitosterolaemia (phytosterolaemia)
Ezetimibe is indicated for the reduction of elevated sitosterol and campesterol levels in patients with homozygous familial sitosterolaemia.

Posology and method of administration
The patient should be on an appropriate lipid-lowering diet and should continue on this diet during treatment with ezetimibe.

Use in Patients with Primary Hypercholesterolemia
The recommended dose of ezetimibe is 10 mg once daily, used alone, with a statin or with fenofibrate. Ezetimibe can be administered at any time of the day, with or without food.

Ezetimibe may be administered with a statin (in patients with primary hypercholesterolemia) or with fenofibrate (in patients with mixed hyperlipidemia) for incremental effect. For convenience, the daily dose of ezetimibe may be taken at the same time as the statin or fenofibrate, according to the dosing recommendations for the respective medications.

If Ezetimibe 10mg tablets are used in combination with a statin therapy, the dosage instructions for that particular statin should be consulted.

When initiating lipid lowering treatment, which includes Ezetimibe 10mg Tablets and a statin in combination, the indicated usual initial dose of that particular statin should be used or the already established higher statin dose should be continued.

If the statin dose is to be increased for the first time or further, the dosage instructions of that particular statin should be followed (such as dose increase only after at least 4 weeks of regular use of the combination without any change).

The stepwise increase of the statin dose in combination treatment results in a relatively small additional decrease of LDL-C, but increases the risk of dose-related adverse events of the statin. This has to be considered for the risk-benefit-assessment when the statin dose is considered.

Use in Patients with Coronary Heart Disease
Combination Therapy with a Statin
For incremental cardiovascular event reduction in patients with coronary heart disease, Ezetimibe 10 mg may be administered with a statin with proven cardiovascular benefit.

Use in the Elderly
No dosage adjustment is required for elderly patients. However, greater sensitivity of some older individuals cannot be ruled out.

Use in Paediatric Patients
Children and adolescents ≥ 10 years: No dosage adjustment is required.
Children <10 years: Treatment with Ezetimibe is not recommended.

Use in Hepatic Impairment
No dosage adjustment is required in patients with mild hepatic insufficiency (Child Pugh score 5 to 6).

Treatment with ezetimibe is not recommended in patients with moderate (Child Pugh score 7 to 9) or severe (Child Pugh score >9) liver dysfunction.

Use in Renal Impairment
No dosage adjustment is required for renally impaired patients.

Co-administration with bile acid sequestrants
Dosing of Ezetimibe should occur either ≥ 2 hours before or ≥ 4 hours after administration of a bile acid sequestrant.

Contraindications
Hypersensitivity to any component of this medication.

When ezetimibe is co-administered with a statin, please refer to the SPC for that particular medicinal product.

Therapy with ezetimibe co-administered with a statin is contraindicated during pregnancy and lactation.

Ezetimibe co-administered with a statin is contraindicated in patients with active liver disease or unexplained persistent elevations in serum transaminases.

Special warnings and precautions for use
When ezetimibe is co-administered with a statin, please refer to the SPC for that particular medicinal product.

Liver enzymes
Co-administration in patients receiving ezetimibe with a statin, consecutive transaminase elevations (≥3 X the upper limit of normal [ULN]) have been observed. When ezetimibe is co-administered with a statin, liver function tests should be performed at initiation of therapy and according to the recommendations of the statin.

Skeletal muscle
Cases of myopathy and rhabdomyolysis have been reported. Most patients who developed rhabdomyolysis were taking a statin concomitantly with ezetimibe. However, rhabdomyolysis has been reported very rarely with ezetimibe monotherapy and very rarely with the addition of ezetimibe to other agents known to be associated with increased risk of rhabdomyolysis. If myopathy is suspected based on muscle symptoms or is confirmed by a creatine phosphokinase (CPK) level >10 times the ULN, ezetimibe, any statin, and any of these other agents that the patient is taking concomitantly should be immediately discontinued. All patients starting therapy with ezetimibe should be advised of the risk of myopathy and told to report promptly any unexplained muscle pain, tenderness or weakness.

Hepatic insufficiency
Due to the unknown effects of the increased exposure to ezetimibe in patients with moderate or severe hepatic insufficiency, ezetimibe is not recommended.

Paediatric Patients
Efficacy and safety of ezetimibe in patients 6 to 10 years of age with heterozygous familial or non-familial hypercholesterolemia have been evaluated. Effects of ezetimibe for treatment periods >12 weeks have not been studied in this age group.

Ezetimibe has not been studied in patients younger than 6 years of age.

Efficacy and safety of ezetimibe co-administered with simvastatin in patients 10 to 17 years of age with heterozygous familial hypercholesterolemia have been evaluated in adolescent boys (Tanner stage II or above) and in girls who were at least one year post-menarche.

There was generally no detectable effect on growth or sexual maturation in the adolescent boys or girls, or any effect on menstrual cycle length in girls. However, the effects of ezetimibe for a treatment period > 33 weeks on growth and sexual maturation have not been studied.

The safety and efficacy of ezetimibe co-administered with doses of simvastatin above 40 mg daily have not been studied in paediatric patients 10 to 17 years of age.

The safety and efficacy of ezetimibe co-administered with simvastatin have not been studied in paediatric patients < 10 years of age.

The long-term efficacy of therapy with ezetimibe in patients below 17 years of age to reduce morbidity and mortality in adulthood has not been studied.

Fibrates
The safety and efficacy of ezetimibe administered with fibrates have not been established.

If cholelithiasis is suspected in a patient receiving ezetimibe and fenofibrate, gallbladder investigations are indicated and this therapy should be discontinued.

Ciclosporin
Caution should be exercised when initiating ezetimibe in the setting of ciclosporin. Ciclosporin concentrations should be monitored in patients receiving ezetimibe and ciclosporin.

Anticoagulants
If ezetimibe is added to warfarin, another coumarin anticoagulant, or fluindione, the International Normalised Ratio (INR) should be appropriately monitored.

Excipient
Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose galactose malabsorption should not take this medicine.

Interaction with other medicinal products and other forms of interaction
There are only information available for adult population. .

It has been shown that ezetimibe does not induce cytochrome P450 drug metabolising enzymes. No clinically significant pharmacokinetic interactions have been observed between ezetimibe and drugs known to be metabolised by cytochromes P450 1A2, 2D6, 2C8, 2C9, and 3A4, or N-acetyltransferase.

Ezetimibe had no effect on the pharmacokinetics of dapsone, dextromethorphan, digoxin, oral contraceptives (ethinyl estradiol and levonorgestrel), glipizide, tolbutamide, or midazolam, during co-administration. Cimetidine, co-administered with ezetimibe, had no effect on the bioavailability of ezetimibe.

Antacids: Concomitant antacid administration decreased the rate of absorption of ezetimibe but had no effect on the bioavailability of ezetimibe. This decreased rate of absorption is not considered clinically significant.

Cholestyramine: Concomitant cholestyramine administration decreased the mean area under the curve (AUC) of total ezetimibe (ezetimibe + ezetimibe glucuronide) approximately 55%. The incremental low-density lipoprotein cholesterol (LDL-C) reduction due to adding ezetimibe to cholestyramine may be lessened by this interaction.

Fibrates: In patients receiving fenofibrate and ezetimibe, physicians should be aware of the possible risk of cholelithiasis and gallbladder disease.

If cholelithiasis is suspected in a patient receiving ezetimibe and fenofibrate, gallbladder investigations are indicated and this therapy should be discontinued.

Concomitant fenofibrate or gemfibrozil administration modestly increased total ezetimibe concentrations (approximately 1.5- and 1.7-fold respectively).

Co-administration of ezetimibe with other fibrates has not been studied.

Fibrates may increase cholesterol excretion into the bile, leading to cholelithiasis. Ezetimibe sometimes increased cholesterol in the gallbladder bile, but not in all species. A lithogenic risk associated with the therapeutic use of ezetimibe cannot be ruled out.

Statins: No clinically significant pharmacokinetic interactions were seen when ezetimibe was co- administered with atorvastatin, simvastatin, pravastatin, lovastatin, fluvastatin or rosuvastatin.

Ciclosporin:
Caution should be exercised when initiating ezetimibe in the setting of ciclosporin. Ciclosporin concentrations should be monitored in patients receiving ezetimibe and ciclosporin.

Anticoagulants: Concomitant administration of ezetimibe (10 mg once daily) had no significant effect on bioavailability of warfarin and prothrombin time in healthy adult males. However, there have been reports of increased International Normalised Ratio (INR) in patients who had ezetimibe added to warfarin or fluindione. If ezetimibe is added to warfarin, another coumarin anticoagulant, or fluindione, INR should be appropriately monitored.

Fertility, pregnancy and lactation
Ezetimibe co-administered with a statin is contraindicated during pregnancy and lactation, please refer to the SPC for that particular statin.

Pregnancy:
Ezetimibe should be given to pregnant women only if clearly necessary. No clinical data are available on the use of ezetimibe during pregnancy. Animal studies on the use of ezetimibe in monotherapy have shown no evidence of direct or indirect harmful effects on pregnancy, embryofoetal development, birth or postnatal development.

Lactation:
Ezetimibe should not be used during lactation. Studies on rats have shown that ezetimibe is secreted into breast milk. It is not known if ezetimibe is secreted into human breast milk.

Fertility:
No clinical data are available on the effects of ezetimibe on human fertility. Ezetimibe had no effect on the fertility of male or female rats.

Effects on ability to drive and use machines
No studies on the effects on the ability to drive and use machines have been performed. However, when driving vehicles or operating machines, it should be taken into account that dizziness has been reported.

Undesirable effects
Ezetimibe administered alone or co-administered with a statin:
Frequencies are defined as very common; common; uncommon; rare, very rare and not known.

PE00000 **MY**

ICONGRAPHICS CODE:

PANTONE SHADE

PANTONE BLACK PROCESS C

Supersedes Artwork Code

PHARMACODE:

SAME SIZE ARTWORK
LEAFLET SIZE : 115 mm x 280 mm

35 mm

Ezetimibe monotherapy		
System organ class	**Adverse reactions**	**Frequency**
Investigations	ALT and/or AST increased; blood CPK increased; gamma-glutamyltransferase increased; liver function test abnormal	Uncommon
Respiratory, Thoracic and Mediastinal Disorders	cough	Uncommon
Gastrointestinal Disorders	abdominal pain; diarrhoea; flatulence	Common
	dyspepsia; gastrooesophageal reflux disease; nausea	Uncommon
Musculoskeletal and Connective Tissue Disorders	arthralgia; muscle spasms; neck pain	Uncommon
Metabolism and Nutrition Disorders	decreased appetite	Uncommon
Vascular Disorders	hot flush; hypertension	Uncommon
General Disorders and Administration Site Condition	fatigue	Common
	chest pain, pain	Uncommon

Additional adverse reactions with ezetimibe co-administered with a statin		
System organ class	**Adverse reactions**	**Frequency**
Investigations	ALT and/or AST increased	Common
Nervous System Disorders	headache	Common
	paraesthesia	Uncommon
Gastrointestinal Disorders	dry mouth; gastritis	Uncommon
Skin and Subcutaneous Tissue Disorders	pruritus; rash; urticaria	Uncommon
Musculoskeletal and Connective Tissue Disorders	myalgia	Common
	back pain; muscular weakness; pain in extremity	Uncommon
General Disorders and Administration Site Condition	asthenia; oedema peripheral	Uncommon

Post-marketing Experience (with or without a statin)		
System organ class	**Adverse reactions**	**Frequency**
Blood and lymphatic system disorders	thrombocytopaenia	Not known
Nervous System Disorders	dizziness; paraesthesia	Not known
Respiratory, thoracic and mediastinal disorders	dyspnoea	Not known
Gastrointestinal disorders	pancreatitis; constipation	Not known
Skin and subcutaneous tissue disorders	erythema multiforme	Not known
Musculoskeletal and connective tissue disorder	myalgia; myopathy/ rhabdomyolysis	Not known
General disorders and administration site conditions	asthenia	Not known
Immune system disorders	hypersensitivity, including rash, urticaria, anaphylaxis and angio-oedema	Not known
Hepatobiliary disorders	hepatitis; cholelithiasis; cholecystitis	Not known
Psychiatric disorders	depression	Not known

Ezetimibe co-administered with fenofibrate: Gastrointestinal disorders: abdominal pain (common).

The elevations (>3 X ULN, consecutive) in serum transaminases and cholecystectomy were noted for fenofibrate monotherapy and ezetimibe co-administered with fenofibrate.

Overdose
Administration of ezetimibe, 50 mg/day, to healthy human for up to 14 days, or 40 mg/day to patients with primary hypercholesterolaemia for up to 56 days, was generally well tolerated. In animals, no toxicity was observed after single oral doses of 5,000 mg/kg of ezetimibe in rats and mice and 3,000 mg/kg in dogs.

A few cases of overdosage with ezetimibe have been reported: most have not been associated with adverse experiences. Reported adverse experiences have not been serious. In the event of an overdose, symptomatic and supportive measures should be employed.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic properties
Pharmacotherapeutic group: Other lipid modifying agents. ATC code: C10A X09

Ezetimibe is in a new class of lipid-lowering compounds that selectively inhibit the intestinal absorption of cholesterol and related plant sterols. Ezetimibe is orally active, and has a mechanism of action that differs from other classes of cholesterol-reducing compounds (e.g. statins, bile acid sequestrants [resins], fibric acid derivatives, and plant stanols). The molecular target of ezetimibe is the sterol transporter, Niemann-Pick C1 Like 1 (NPC1L1), which is responsible for the intestinal uptake of cholesterol and phytosterols.

Ezetimibe localises at the brush border of the small intestine and inhibits the absorption of cholesterol, leading to a decrease in the delivery of intestinal cholesterol to the liver; statins reduce cholesterol synthesis in the liver and together these distinct mechanisms provide complementary cholesterol reduction.

Ezetimibe inhibited the absorption of [^{14}C]-cholesterol with no effect on the absorption of triglycerides, fatty acids, bile acids, progesterone, ethinyl estradiol, or fat soluble vitamins A and D.

Cardiovascular morbidity and mortality vary directly with the level of total-C and LDL-C and inversely with the level of HDL-C.

Administration of Ezeimibe with a statin is effective in reducing the risk of cardiovascular events in patients with coronary heart disease and ACS event history.

Ezetimibe either as monotherapy or coadministered with a statin significantly reduced total cholesterol (totalC), lowdensity lipoprotein cholesterol (LDLC), apolipoprotein B (Apo B),and triglycerides (TG) and increased highdensity lipoprotein cholesterol (HDLC) in patients with hypercholesterolaemia.

Pharmacokinetic properties
Absorption: After oral administration, ezetimibe is rapidly absorbed and extensively conjugated to a pharmacologically active phenolic glucuronide (ezetimibe glucuronide). Mean maximum plasma concentrations (C_{max}) occur within 1 to 2 hours for ezetimibe-glucuronide and 4 to 12 hours for ezetimibe. The absolute bioavailability of ezetimibe cannot be determined as the compound is virtually insoluble in aqueous media suitable for injection.

Concomitant food administration (high fat or non-fat meals) had no effect on the oral bioavailability of ezetimibe when administered as ezetimibe 10-mg tablets. Ezetimibe can be administered with or without food.

Distribution: Ezetimibe and ezetimibe-glucuronide are bound 99.7% and 88 to 92% to human plasma proteins, respectively.

Biotransformation: Ezetimibe is metabolised primarily in the small intestine and liver via glucuronide conjugation with subsequent biliary excretion. Minimal oxidative metabolism has been observed in all species. Ezetimibe and ezetimibe- glucuronide are the major drug-derived compounds detected in plasma, constituting approximately 10 to 20 % and 80 to 90 % of the total drug in plasma, respectively. Both ezetimibe and ezetimibe- glucuronide are slowly eliminated from plasma with evidence of significant enterohepatic recycling. The half-life for ezetimibe and ezetimibeglucuronide is approximately 22 hours.

Elimination: Following oral administration of ^{14}C-ezetimibe (20 mg), total ezetimibe accounted for approximately 93% of the total radioactivity in plasma. Approximately 78% and 11% of the administered radioactivity were recovered in the faeces and urine, respectively, over a 10-day collection period. After 48 hours, there were no detectable levels of radioactivity in the plasma.

Special populations:

Paediatric patients
The pharmacokinetics of ezetimibe are similar between children ≥6 years and adults. Pharmacokinetic data in the paediatric population <6 years of age are not available.

Geriatric patients
Plasma concentrations for total ezetimibe are about 2-fold higher in the elderly (≥65 years) than in the young (18 to 45 years). LDL-C reduction and safety profile are comparable between elderly and young subjects treated with ezetimibe. Therefore, no dosage adjustment is necessary in the elderly.

Hepatic insufficiency
No dosage adjustment is necessary for patients with mild hepatic insufficiency. Due to the unknown effects of the increased exposure to ezetimibe in patients with moderate or severe (Child Pugh score >9) hepatic insufficiency, Ezetimibe is not recommended in these patients.

Renal insufficiency
No dosage adjustment is necessary for renally impaired patients.

Patient (post-renal transplant and receiving multiple medications, including ciclosporin) had a 12-fold greater exposure to total ezetimibe.

Gender
Plasma concentrations for total ezetimibe are slightly higher (approximately 20%) in women than in men. LDL-C reduction and safety profile are comparable between men and women treated with ezetimibe. Therefore, no dosage adjustment is necessary on the basis of gender.

PHARMACEUTICAL PARTICULARS

List of excipients:
Lactose Monohydrate, Sodium Lauryl Sulphate, Croscarmellose Sodium, Povidone K-30, Magnesium Stearate

Incompatibilities
Not Applicable

Shelf life
24 Months

Special precautions for storage
Store below 30°C, Protect from Moisture

Nature and contents of container
Blister pack of PVC/Aclar clear film and plain aluminium foil.

Pack Size: 3x10's

DATE OF REVISION OF THE TEXT
February 2021

Product Registration Holder :
Glenmark Pharmaceuticals
(Malaysia) Sdn Bhd (660397-W)
D-31-02, Menara Suezcap 1,
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Manufactured by:
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Indore Special Economic Zone,
Pithampur, Dist. Dhar (M.P),
Pincode: 454775, India.
TM Trade Mark

PE0000 **MY**

ICONGRAPHICS CODE:

PANTONE SHADE

PANTONE BLACK PROCESS C

Supersedes Artwork Code

PHARMACODE: